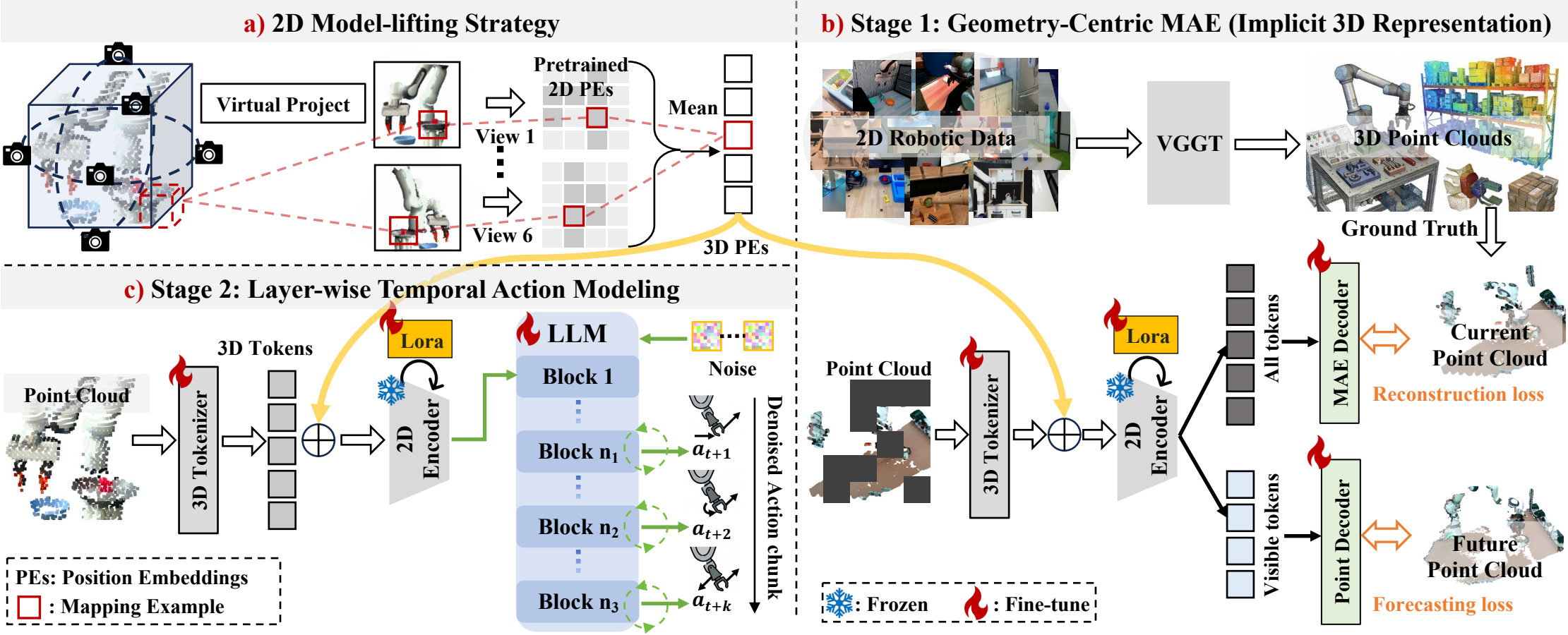
a) 2D Model-lifting Strategy
Virtual Project
Pretrained 2D PEs
View 1
View 6
Mean
3D PEs
b) Stage 1: Geometry-Centric MAE (Implicit 3D Representation)
2D Robotic Data
VGGT
3D Point Clouds
Ground Truth
Point Cloud
3D Tokenizer
Lora
2D Encoder
All tokens
Visible tokens
MAE Decoder
Point Decoder
Current Point Cloud
Reconstruction loss
Future Point Cloud
Forecasting loss
: Frozen
: Fine-tune
c) Stage 2: Layer-wise Temporal Action Modeling
Point Cloud
3D Tokenizer
3D Tokens
Lora
2D Encoder
LLM
Block 1
Block n1
Block n2
Block n3
Noise
Denoised Action chunk
PEs: Position Embeddings
: Mapping Example